AudioCleanUpTool

Crisp — Audio Cleanup for Everyone

User Guide · v0.1.0

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1. Introduction

AudioCleanUpTool (code name: *Crisp*) is a free, open-source desktop application for cleaning and polishing audio recordings. Whether you record podcasts, church services, voice-overs, or anything in between, Crisp gives you professional results without needing to know anything about audio engineering.

The entire processing pipeline runs locally on your machine — no cloud, no subscription, no internet connection required. Your audio never leaves your computer.

1.1 Who is this for?

- Podcasters who want clean, loud, professional-sounding episodes
- Church AV teams cleaning up recordings of sermons and services
- Voice-over artists preparing recordings to ACX or broadcast spec
- Anyone who records speech and wants it to sound better quickly
- Developers who want a scriptable Python audio pipeline

1.2 Key principles

- **Non-destructive:** the original file is never overwritten — you always export a new copy.
- **Transparent:** every processing step is visible, adjustable, and can be toggled off.
- **A/B comparison:** switch between the original and cleaned version at any time to judge the difference.
- **One-click option:** Auto Clean analyses your audio and picks the right settings automatically.

2. Installation

2.1 Windows installer (recommended)

Download **AudioCleanUpTool-0.1.0-Setup.exe** from the Releases page on GitHub and run it. The installer:

- Installs the app to *C:\Program Files\AudioCleanUpTool* (or a folder of your choice)
- Creates a Start Menu entry
- Optionally creates a Desktop shortcut
- Bundles Python, all dependencies, and ffmpeg — nothing else to install

■■ *No administrator rights are required. The installer defaults to a per-user installation.*

2.2 Running from source (developers)

Requires Python 3.10 or later.

```
git clone https://github.com/mourninggrace/crisp-audio.git cd crisp-audio python
-m venv .venv .venv\Scripts\Activate.ps1 # PowerShell pip install -e .[dev] python
```

```
-m crisp
```

2.3 Building the executable yourself

```
pip install -e .[dev] pyinstaller crisp.spec # Produces dist\Crisp.exe &
"C:\Program Files (x86)\Inno Setup 6\ISCC.exe" installer\crisp.iss # Produces
dist\AudioCleanUpTool-0.1.0-Setup.exe
```

3. Getting Started

3.1 The interface at a glance

The window is divided into three areas:

Area	Location	What it does
Left panel	Left column	Source input, cleanup processors, export controls
Waveform area	Centre	Before/after waveform display + playback transport
Menu bar	Top	File, Edit, View, Settings, Help
Status bar	Bottom	Live status messages and progress feedback

■■ *All three left-panel sections (Source, Cleanup, Export) are resizable — drag the dividers between them.*

3.2 Quick start: clean a file in 4 steps

Step	Action
1	Click Open file… and choose any WAV, FLAC, MP3, AAC, OGG, or AIFF file.
2	Click ■ Auto Clean. Crisp analyses the audio and applies the right settings automatically.
3	Use the A · Original / B · Cleaned buttons to compare.
4	Choose an export preset and click Export cleaned….

3.3 Recording audio directly

Instead of opening a file, you can record from a microphone:

- Select your input device from the **Input device** drop-down in the Source section.
- Click ● **Record**. The level meter shows the incoming signal.
- Click ■ **Stop** when done. The recording appears in the waveform view automatically.
- Proceed with Auto Clean or manual cleanup as normal.

4. Audio Processors

Crisp has 10 processing stages, each with its own on/off toggle and parameter controls. They are organised into five tabs in the Cleanup section.

Processing runs in a fixed pipeline order that is designed to give the best results: noise removal first, then reverb, plosives, EQ, clarity, loudness, trim, then effects.

Tab 1 — Noise Reduction

Noise Reduction

Removes constant background noise: hum, hiss, fan noise, air conditioning, electrical interference. Uses spectral gating — frequencies that are consistently present at low levels are identified as noise and suppressed.

Parameter	Range	Default	What it does
Strength	0.0 – 1.0	0.85	How aggressively noise is reduced. Higher = more reduction, but too high
Constant noise floor	On/Off	Off	On: assumes the noise floor is constant throughout (best for steady hum/

■■ *Start with default settings. Only increase Strength if the noise is clearly still audible after processing.*

Plosive Removal

Removes plosives — the sharp low-frequency 'pop' caused by breath hitting the microphone on P, B, and T sounds. Detects frames where sub-120 Hz energy spikes well above the rolling average and ducks the low band at those moments, leaving normal speech untouched.

Parameter	Range	Default	What it does
Crossover	60 – 200 Hz	120 Hz	Frequency below which energy is monitored for pops.
Sensitivity	1.5× – 6.0×	3.0×	How much louder than average the low end must be to count as a plosive
Reduction	0.0 – 1.0	0.85	How much of the low-band energy to remove at detected plosives.

Tab 2 — Dereverb

Reduce reverb / room sound

Suppresses the diffuse room tail — the way sound bounces around a room and makes recordings sound distant or echoey. Uses spectral over-subtraction on the slowly-varying energy floor. Works well on speech recorded in untreated rooms. This stage is the most likely to colour the sound, so it's off by default — enable it when the room is clearly audible.

Parameter	Range	Default	What it does
Strength	0.0 – 1.0	0.60	How much of the room tail to remove.
Tail smoothing	20 – 200 ms	60 ms	Smoothing window for the reverb tail estimate. Longer = smoother but slo

Tab 3 — EQ & Clarity

Balance EQ

Corrective EQ for spoken voice. Applies three gentle adjustments:

- **Rumble cut:** high-pass filter that removes sub-sonic low-frequency content (mic handling noise, HVAC rumble, vibration). Adjustable from 40–150 Hz.
- **De-box:** a small dip around 300 Hz — the 'boxy' or 'muddy' frequency range where many untreated rooms and budget microphones build up. Usually –2 to –3 dB.
- **Air:** a gentle boost around 12 kHz adds sparkle and openness. Usually 1–3 dB.

Parameter	Range	Default	What it does
Rumble cut	40 – 150 Hz	80 Hz	High-pass filter cutoff frequency.
De-box	–6 – 0 dB	–2.5 dB	Reduction at ~300 Hz.
Air	0 – 6 dB	+2.0 dB	Boost at ~12 kHz.

Enhance voice clarity

Lifts the 2–5 kHz presence range — where consonants and speech intelligibility live — and optionally runs a gentle compressor to bring quiet consonants forward without letting louder vowels clip. Run after noise and reverb are dealt with so you're not amplifying artefacts.

Parameter	Range	Default	What it does
Presence freq	2000 – 6000 Hz	3500 Hz	Centre of the presence boost.
Presence lift	0 – 6 dB	+3.0 dB	Amount of boost.
Gentle leveling	On/Off	On	Enables the soft compressor for even dynamics.
Leveling ratio	1.5 – 4.0:1	2.5:1	Compression ratio. Higher = more evening-out.

Tab 4 — Loudness

Normalize loudness (LUFS)

Measures integrated loudness using the ITU-R BS.1770 standard (LUFS — Loudness Units relative to Full Scale) and applies the exact gain needed to hit the target. Then guards the output with a peak ceiling to prevent clipping. This is the last corrective stage before effects.

Parameter	Range	Default	What it does
Target	–30 – –9 LUFS	–16 LUFS	Target integrated loudness. Common values: –14 (YouTube/Spotify), –16 (
Peak ceiling	–3 – 0 dBFS	–1 dBFS	Maximum allowed sample peak after gain is applied.

■■ *The current LUFS reading for the loaded clip is shown in the bottom-right of the waveform area, next to the A/B buttons.*

Tab 5 — Effects

Trim (gain)

Simple dB gain adjustment. Useful when the loudness normaliser can't bring the level up enough on its own, or when you want to manually reduce gain before other stages. Clips output to ±1.0 (0 dBFS) to prevent overflow.

Parameter	Range	Default	What it does
Gain	–24 – +24 dB	0 dB	Gain adjustment. Positive = louder, negative = quieter.

Chorus

Thickens audio by layering multiple LFO-modulated delay copies of the signal over the dry signal. Each voice uses a slightly different LFO phase offset, giving a lush ensemble effect. Useful for music and creative voice processing.

Parameter	Range	Default	What it does
Rate	0.1 – 5.0 Hz	0.8 Hz	LFO modulation speed. Faster = more movement.
Depth	0.001 – 0.030 s	0.010 s	LFO delay swing. More depth = wider modulation.
Mix	0.0 – 1.0	0.40	Wet/dry blend. 0 = dry only, 1 = fully wet.
Voices	1 – 3	2	Number of chorus voices. More voices = thicker but heavier CPU.

Widener

Applies M/S (mid-side) stereo widening. Increases or decreases the width of the stereo image without affecting mono compatibility. Mono input is automatically upmixed to stereo before widening.

Parameter	Range	Default	What it does
Width	0.0 – 2.0	1.0	0 = mono (all mid), 1 = unchanged, >1 = wider stereo image.

■■ *Width values above 1.5 can cause some elements to become very wide or phase-y on speakers. Use on headphone mixes cautiously.*

Panner

Positions the signal in the stereo field using a constant-power pan law (sin/cos) so perceived loudness stays stable as you pan. Mono input is upmixed to stereo before panning. Output is always stereo.

Parameter	Range	Default	What it does
Pan position	0.0 – 1.0	0.5	0.0 = hard left, 0.5 = centre, 1.0 = hard right.

5. Auto Clean

Auto Clean is the one-button shortcut. It analyses your audio across six dimensions and automatically configures and applies the most appropriate settings. It's a good starting point even if you plan to tweak afterwards — the analysis report tells you exactly what it measured and what it applied.

5.1 What Auto Clean measures

Measurement	What it is
Content type	Speech/voice vs music/instrument, using autocorrelation pitch detection.
Loudness	Integrated loudness in LUFS (ITU-R BS.1770).
Estimated SNR	Signal-to-noise ratio using Welch PSD + minimum noise floor tracking.
Reverb tail	RT60 estimate — how long the room echo takes to decay by 60 dB.
Plosives	Whether breath pops are present in the low-frequency band.
Spectral tilt	Whether the audio is bright or dull/muffled (high vs low-mid energy ratio).

5.2 What happens after analysis

- A settings configuration is built automatically based on the measurements.
- The processor panels in the Cleanup section update to show the applied settings.
- The cleanup pipeline runs in the background.
- When done, a report dialog summarises the measurements and what was applied.
- You can switch to B (Cleaned) to hear the result, then tweak any settings manually and re-apply if needed.

5.3 When to use Auto Clean vs manual

Situation	Recommendation
Quick turnaround, don't want to think about audio	Auto Clean — done.
Good recording, just needs loudness normalising	Manual: enable Loudness only, pick a preset target.
Very reverberant room	Auto Clean, then increase Dereverb strength if needed.
Music recording (not voice)	Auto Clean detects music and adjusts; or manual with effects.
Specific platform target (e.g. Apple Podcasts)	Auto Clean, then use Export Presets for the correct LUFS/format.

6. A/B Comparison & Waveform View

6.1 The waveform display

The centre panel shows two waveforms side by side:

- **A (top):** the original, unprocessed audio.
- **B (bottom):** the cleaned/processed audio (appears after cleanup runs).

Both waveforms are x-linked — zooming or scrolling one scrolls both, so you can compare the same section of audio at the same position.

6.2 Playback and A/B toggle

Control	Action
■ Play / ■ Stop	Start or stop playback of the selected version.
A · Original	Switch playback to the original recording. The live LUFS reading updates.
B · Cleaned	Switch playback to the processed result. The live LUFS reading updates.
LUFS: — (label)	Shows the current integrated loudness of whichever version is selected.

■■ *Switching A/B while playing takes effect immediately — no need to stop first.*

6.3 The playhead

A vertical line follows playback position in real time across both waveforms. Playback stops automatically when the clip ends.

7. Exporting Audio

7.1 Single file export

- Choose a **Preset** (or leave on Custom for manual control).
- Choose a **Format**.
- For lossless formats (WAV/FLAC/AIFF), choose a **Bit depth**.
- For lossy formats (MP3/AAC/OGG/OPUS), choose a **Bitrate**.
- Click **Export cleaned…** and choose where to save.

If cleanup has been applied, the cleaned audio is exported. If not, the original is exported.

7.2 Export presets

Preset	Format	LUFS Target	Bit depth / Bitrate	Use case
YouTube	MP3	–14	192 kbps	Video uploads
Spotify	MP3	–14	320 kbps	Music streaming
Podcast (general)	MP3	–16	128 kbps	Podcast hosts
Apple Podcasts	AAC	–16	128 kbps	Apple Podcasts spec

Broadcast (EBU R128)	WAV	–23	24-bit	European broadcast
Broadcast TV (ATSC A/85)	WAV	–24	24-bit	US broadcast TV
Church / Livestream	AAC	–16	256 kbps	Live stream uploads
Voice-Over / ACX	MP3	–20	192 kbps	Audiobooks / ACX
Archival (lossless)	FLAC	none	24-bit	Master archive

7.3 Supported formats

Format	Extension	Type	Notes
WAV	.wav	Lossless	PCM 16-bit, 24-bit, or 32-bit float. Universal compatibility.
FLAC	.flac	Lossless	Compressed lossless. Smaller than WAV, bit-perfect.
MP3	.mp3	Lossy	Universal. 128–320 kbps. Requires bundled ffmpeg.
AAC	.m4a	Lossy	Better quality than MP3 at same bitrate. Apple/YouTube standard.
OGG	.ogg	Lossy	Open format. Good quality. Widely supported on Linux/web.
OPUS	.opus	Lossy	Excellent quality at low bitrates. Great for voice. Via ffmpeg.
AIFF	.aiff	Lossless	Uncompressed. Mac-friendly equivalent of WAV.

■■ *ffmpeg is bundled with the installer — no separate download needed for MP3, AAC, or OPUS export.*

7.4 Batch processing

Batch processing runs the entire cleanup pipeline over a folder of audio files and exports them all at once:

- Click **Batch folder…** in the Export section.
- Select the folder of source audio files.
- Select an output folder.
- Crisp processes every supported audio file using the currently configured cleanup settings.
- A summary dialog reports how many files were processed successfully.

■■ *The cleanup settings used for batch processing are whatever is currently configured in the Cleanup panel — the same settings you'd use for a single file. Set them up first, then batch.*

8. Settings & Preferences

8.1 Saving and loading cleanup settings

Your processor settings (which stages are enabled and all parameter values) can be saved and reloaded as **.crisp.json** files. This lets you build preset configurations for different recording scenarios:

- Click **Save settings…** in the Cleanup section to save the current configuration.

- Click **Load settings…** to load a saved configuration. All processor panels update immediately.
- These files are plain JSON — you can edit them manually or share them with others.

8.2 Preferences dialog (Ctrl+,)

Open **Settings** → **Preferences…** or press **Ctrl+,** to access:

- **Colour theme:** choose from 7 themes. The theme is also accessible from View → Colour theme.
- **Playback volume:** adjust the monitoring level.

8.3 Colour themes

Theme	Description
Dark Default	Deep navy with cyan accents. The default.
Midnight Blue	Deep blue-black with blue highlights.
Charcoal Orange	Dark charcoal with orange accent colour.
Deep Purple	Dark purple palette with lavender highlights.
Hacker Green	Black with terminal green. For those of a certain disposition.
Light Clean	White/grey light theme with blue accents.
Warm Cream	Warm off-white light theme with earthy tones.

Theme selection is saved automatically and restored next time you open the app.

9. Keyboard Shortcuts

Shortcut	Action
Ctrl+O	Open audio file
Ctrl+E	Export cleaned audio
Ctrl+,	Open Preferences dialog
Ctrl+Q	Quit the application

10. Tips & Best Practices

10.1 Signal chain order

Crisp always runs processors in the same fixed order, regardless of tab order. This order is designed to give the best results:

Order	Processor	Why this position

1	Noise Reduction	Remove broadband noise first so later stages don't amplify it.
2	Dereverb	Remove room before EQ so you're not EQ-ing the room character.
3	Plosive Removal	Duck plosives before clarity boost to avoid amplifying pops.
4	EQ Balance	Corrective EQ on a now-clean signal.
5	Voice Clarity	Enhance presence and dynamics on a clean, EQ'd signal.
6	Loudness	Normalise loudness after all colouring is done.
7	Trim	Fine-tune gain if needed after normalisation.
8	Chorus	Creative effects go last.
9	Widener	Width after everything else is balanced.
10	Panner	Final stereo placement.

10.2 Common scenarios

Noisy room recording (fan/AC noise)

- Enable Noise Reduction. Start at Strength 0.85.
- If artefacts (musical noise) appear, reduce Strength slightly.
- Enable Constant noise floor if the noise is steady throughout.

Echoey/reverberant room

- Enable Dereverb. Start at Strength 0.6.
- Increase gradually — more strength removes more room but can also affect voice quality.
- Dereverb works best on voice, less well on music.

Plosive microphone pops

- Enable Plosive Removal.
- If pops remain, lower Sensitivity (try 2.0×) to detect more aggressively.
- If normal speech is being affected, raise Sensitivity (try 4.0×).

Preparing for a podcast platform

- Run Auto Clean, or enable noise/plosives/EQ/clarity manually.
- Enable Loudness, set target to –16 LUFS.
- Choose the Podcast (general) export preset for correctly-targeted MP3.

Church livestream recording

- Use Auto Clean for the initial pass.
- Choose the Church / Livestream export preset (–16 LUFS, 256 kbps AAC).
- For batch processing multiple service recordings, use Batch folder….

10.3 LUFS reference targets

Platform / Standard	Target LUFS
YouTube	–14 LUFS integrated
Spotify	–14 LUFS integrated
Apple Music	–16 LUFS integrated
Apple Podcasts	–16 LUFS integrated
General podcasts	–16 LUFS integrated
Voice-over / ACX	–18 to –23 LUFS (aim for –20)
EBU R128 (broadcast)	–23 LUFS integrated
ATSC A/85 (US TV)	–24 LUFS integrated

11. Troubleshooting

Problem	Solution
App doesn't launch	Make sure you ran the full installer, not just Crisp.exe directly. Try right-clicking and running as admini
No input devices listed	Check your microphone is plugged in and Windows has given Crisp permission to access it (Settings
Export failed	MP3/AAC/OPUS export requires ffmpeg, which is bundled in the installer. If running from source, insta
Audio sounds worse after processing	Reduce Noise Reduction strength. Too-aggressive denoising causes 'watery' artefacts. Also check De
Loudness normalisation didn't work	The clip must be at least 400 ms long for LUFS measurement to work. Very short clips are left at their
Batch processing skipped some files	Only WAV, FLAC, OGG, AIFF, MP3, M4A/AAC files are processed. The summary dialog shows how
Settings file won't load	The .crisp.json file must have been saved by Crisp. Processor keys not matching any known processo

12. Open Source & Licence

AudioCleanUpTool is free and open-source software, released under the **MIT License**. You are free to:

- Use it for any purpose, including commercial work
- Modify the source code
- Distribute copies or modified versions
- Include it in other projects

The only requirement is that the copyright notice and licence text are included in any copies or substantial portions of the software.

Source code

GitHub: **https://github.com/mourninggrace/crisp-audio**

Built with

Library	Licence	Purpose
PySide6	LGPL v3	Qt GUI framework
scipy	BSD	DSP filters and signal processing
numpy	BSD	Array operations
noisereduce	MIT	Spectral gating noise reduction
pyloudnorm	MIT	LUFS measurement (ITU-R BS.1770)
soundfile	BSD	WAV/FLAC/AIFF/OGG read/write
sounddevice	MIT	Real-time audio recording
pydub	MIT	MP3/AAC encoding via ffmpeg
imageio-ffmpeg	BSD	Bundled ffmpeg binaries
pyqtgraph	MIT	Waveform display